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Mineralogical and elemental composition of fly ash from pilot scale fluidised bed combustion of lignite, bituminous coal, wood chips and their blends

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Abstract

The chemical and mineralogical composition of fly ash samples collected from different parts of a laboratory and a pilot scale CFB facility has been investigated. The fabric filter and the second cyclone of the two facilities were chosen as sampling points. The fuels used were Greek lignite (from the Florina basin), Polish coal and wood chips. Characterization of the fly ash samples was conducted by means of X-ray fluorescence (XRF), inductive coupled plasma-optical emission spectrometry (ICP-OES), thermogravimetric analysis (TGA), particle size distribution (PSD) and X-ray diffraction (XRD). According to the chemical analyses the produced fly ashes are rich in CaO. Moreover, SiO_2 is the dominant oxide in fly ash with Al_2O_3 and Fe_2O_3 found in considerable quantities. Results obtained by XRD showed that the major mineral phase of fly ash is quartz, while other mineral phases that are occurred are maghemite, hematite, periclase, rutile, gehlenite and anhydrite. The ICP-OES analysis showed rather low levels of trace elements, especially for As and Cr, in many of the ashes included in this study compared to coal ash from fluidised bed combustion in general.

Keywords: CFB fly ash; Bituminous coal; Wood chips; Lignite

1. Introduction

Despite the fact that combustion of solid fuels using conventional combustion technologies is and will probably continue to be an important part of the heat and power generation systems, combustion of solid fuels applying more environmental friendly technologies, such as the circulated fluidised bed (CFB) technology, continuously gains ground. Moreover, this technology has another important advantage in comparison with the conventional technologies; it is able to burn low quality fuels, such as lignite, alternative fuels such as biomass (wood chips) as well as blends of these fuels.

Fly ash can be either an industrial waste material and ecological nuisance, or a valuable raw material. For the latter purpose, its properties need to be defined precisely and controlled so that a uniform and reproducible material can be supplied. This paper describes a comprehensive study of some fly ash samples, with the intention of elucidating their chemical, physical, mineralogical and technical properties as an incentive to their utilization.

A result of the continuous development of the CFB technologies is that the amounts of combustion by-products such as fly ash are steadily increasing [1]. This material has found limited applications in the cement and concrete industry and in land reclamation and restoration.

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The characteristics of fly ash such as type of particles, size distribution and chemical composition are of paramount importance with a view to their potential applications. Fly ash particles may differ, depending on the mineral matter content of the coal and on the way the combustion process is conducted [2].

2. Materials and analytical methods

2.1. Sampling

The fly ash samples examined were produced from the combustion of different solid fuels in laboratory and pilot scale CFB boilers. Investigation has been accomplished through XRF, PSD, XRD and ICP.

The combustion experiments were carried out at the 50 kW lab scale boiler of VTT Technical Research Centre of Finland, and the 12 MW pilot scale boiler of the Chalmers University of Technology (CUT) in Sweden. The experimental facilities are following the flow chart as presented in Fig. 1.

2.2. Analytical methods

The experiment was carried out using Greek lignite, a Polish bituminous coal and wood chips from Swedish conifers, and various blends of these three feeds.

Combustion was accomplished at a temperature of 900 °C in both facilities without the use of limestone.

The proximate and ultimate analyses of the used fuels, as well as their heating values are presented in Table 1.

Table 1
Average characteristics and energy contents of the feed fuels

Parameter	Polish bituminous coal	Greek lignite	Wood chips
Proximate analysis (wt% dry basis)			
Moisture	1.9	36.5	4.6
Volatiles	29.33	41.67	80.10
Ash	12.96	38.36	1.14
Fixed carbon	57.71	19.97	18.76
Ultimate analysis (wt% dry basis)			
Carbon	68.42	32.29	48.77
Hydrogen	3.91	3.42	5.85
Nitrogen	1.32	1.33	0.45
Oxygen	12.69	17.96	43.79
Sulphur	0.7	1.64	0
Gross heating value d.b. (kJ/kg)	29,171	17,887	25,154
Net heating value d.b. (kJ/kg)	28,330	17,150	23,899

During the above mentioned experiments, fly ash samples from the same part of each facility (the fabric filter and the second cyclone) were collected. These samples, the feed fuels combusted and the sampling point are stated in Table 2.

2.2.1. Bulk chemical analysis

The major elements of the solid residues were determined with a Spectro X-Lab 2000 Energy Dispersive X-ray fluorescence (XRF) spectrometer applying the samples in a pressed powder form.

Loss-on-ignition (LOI) tests were performed as follows: moisture, volatile matter, ash and fixed carbon measure-

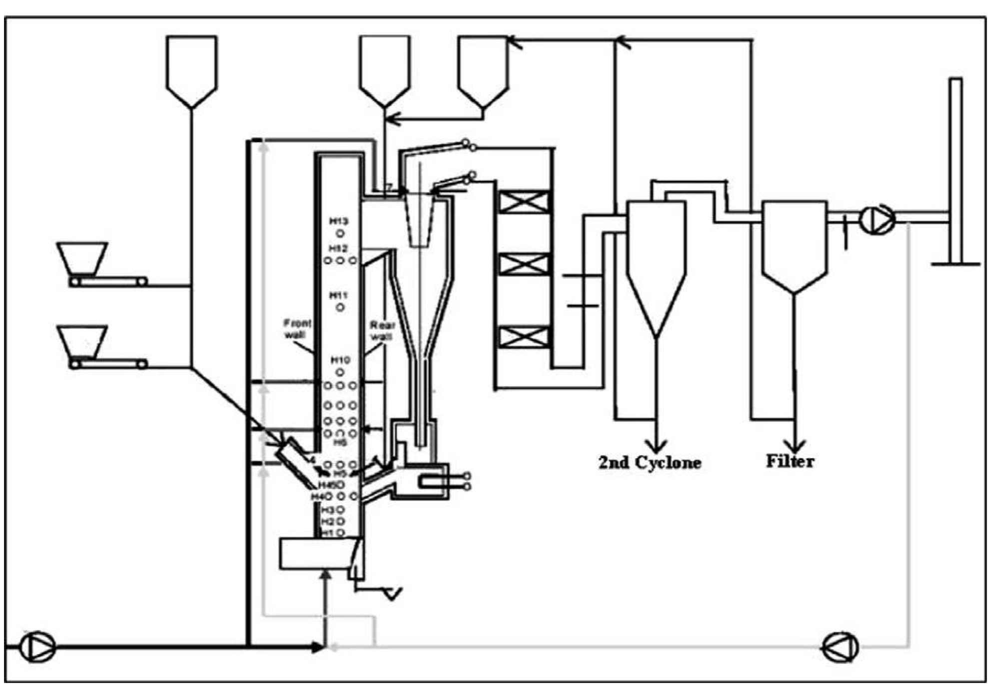

Fig. 1. Schematic diagram of the fluidised bed set-up.

Table 2
Classification of the fly ash samples collected at VTT's and CUT's facility

Sample	Composition of feed fuels	Sampling point/facility	$Al_2O_3 + SiO + Fe_2O_3$	CaO
F1	100% Greek lignite	Fabric filter/VTT	70.03	12.53
F2	50% Wood chips–50% Polish coal	Fabric filter/VTT	62.65	12.68
F3	100% Wood chips	Fabric filter/VTT	56.85	15.04
F4	100% Polish coal	Fabric filter/CUT	45.11	8.67
F5	50% Polish coal–50% Wood chips	Fabric filter/CUT	41.57	18.13
F6	100% Wood chips	Fabric filter/CUT	18.46	41.70
C1	100% Greek lignite	Second cyclone/VTT	80.98	5.73
C2	50% Wood chips–50% Polish coal	Second cyclone/VTT	50.86	14.19
C3	100% Wood chips	Second cyclone/VTT	52.72	19.59
C4	100% Polish coal	Second cyclone/CUT	27.76	9.58
C5	50% Polish coal–50% Wood chips	Second cyclone/CUT	33.22	13.76
C6	100% Wood chips	Second cyclone/CUT	31.69	41.16

ments were conducted by a LECO TGA 501 Thermogravimetric Analyzer. The results were obtained on a dry basis. 2.5 g of each sample was heated from room temperature till 850 °C. During the first minutes of each test, the sample was heated till 107 °C at a rate of 15 °C/min in a nitrogen gas flow. At this temperature the moisture content was received. After the moisture is determined, the chamber was heated at a rate of 25 °C/min until the sample reached 550 °C. At this temperature data for the volatile matter was obtained. After that the temperature raised at 35 °C/min till 850 °C where the ash content of the sample was received.

For the determination of major and trace elements a microwave-assisted acid digestion (MW-AD) followed by an inductively coupled plasma-optical emission spectrometry (ICP-OES) was applied. A combination of HNO_3, HCl and HF is employed in the MW-AD followed by H_3BO_3 addition, for the removal of HF from the reaction mixture.

The microwave digestion was carried out by using a CEM MDS 2000 oven. ICP-OES measurement was performed using a Perkin–Elmer Optima 4300 DV.

The particle size of the ashes was measured with a Malvern Mastersizer-S particle size analyzer using the wet dispersion method in water.

The mineralogical composition of the fly ash samples was examined by XRD using a Siemens D-500 spectrometer with copper Kα radiation.

The chemical and mineralogical analyses of the fly ash samples were carried out at the laboratories of the Centre for Research and Technology Hellas (CERTH).

3. Results and discussion

3.1. Classification of the fly ash samples

Based on the ASTM standards fly ash is classified according to the content of its major elements (Si, Al, Fe and Ca). ASTM C 618 [3] defines three classes:

- Class N which includes raw of calcinated natural pozzolans with at least 70% SiO_2, Al_2O_3 and Fe_2O_3.
- Class F comprising ash produced from anthracite or bituminous coal combustion with at least 70% SiO_2, Al_2O_3 and Fe_2O_3.
- Class C comprising ash produced from lignite or sub-bituminous coal combustion with at least 50% but less than 70% SiO_2, Al_2O_3 and Fe_2O_3.

Class F fly ashes usually contain less than 5% CaO, while fly ashes belonging to class C contain a large proportion of CaO (10–35%).

From the chemical point of view the analysed ash samples are rich in CaO, while the sum of the oxides Al_2O_3, SiO_2 and Fe_2O_3 is over 50% but less than 70%. This sum as well as the concentration of CaO is shown in Table 2. According to the ASTM standard, the fly ashes are classified as class C, except of sample C1 (lignite), which is associated with class F. It is also worthwhile to mention that the bituminous coal produces fly ash of class C.

3.2. Unburned carbon content

The loss-on-ignition (LOI), which measures the amount of the unburned carbon remaining in the fly ash, is one of the most significant chemical properties of the samples analysed. The maximum LOI determined in these experiments, shown in Table 3, was F4 (coal) (32.93%) and C4 (coal) (53.68%). The ASTM C-618 [3] standard has an upper limit of 6%, concerning class C fly ash utilization as a pozzolan in Portland cement. In general, low-carbon ashes are required because carbon may interfere with air entrainment by adsorbing entrainment additives, although the carbon content does not always have this effect [4]. According to the aforementioned specification, good results are obtained by the fly ash samples F1 (lignite), F6 (wood chips) and C1 (lignite), C3 (wood chips). In compliance with ASTM C-618 [3], SO_3 content should be lower than 5%. In this case only the sample C1 (lignite) from the second cyclone meets this requirement.

Table 3
Ash composition (%) of the samples as determined by XRF analysis

	F1	F2	F3	F4	F5	F6	C1	C2	C3	C4	C5	C6
SiO_2	40.33	36.58	32.96	23.35	22.81	13.39	47.80	28.05	29.20	14.57	18.72	20.07
Fe_2O_3	8.11	6.53	6.01	6.34	4.69	2.46	9.06	7.70	8.89	4.74	5.53	2.36
Al_2O_3	21.59	19.54	17.88	15.42	14.07	2.61	24.12	15.11	14.63	8.45	8.97	9.26
TiO_2	1.34	1.06	0.90	0.77	0.69	0.03	0.98	0.48	0.39	0.29	0.27	0.05
CaO	12.53	12.68	15.04	8.67	18.13	41.70	5.73	14.19	19.59	9.58	13.76	41.16
MgO	4.10	3.58	4.18	4.71	5.81	8.10	3.64	5.00	5.70	2.99	3.89	4.47
SO_3	8.09	7.51	8.98	3.37	5.86	10.08	3.80	9.65	13.23	3.82	4.81	4.02
P_2O_5	0.71	0.92	1.37	0.87	2.55	4.67	0.50	0.73	1.03	0.31	0.78	2.34
Na_2O	0.30	0.63	1.23	2.17	0.88	2.75	0.28	1.46	1.95	1.10	1.82	0.63
K_2O	2.25	2.05	2.26	1.65	6.41	12.08	3.13	2.23	2.96	0.91	3.07	5.57
LOI	1.10	9.34	9.51	32.93	18.15	2.17	1.19	16.24	2.64	53.68	33.36	10.51
Total	100.35	100.42	100.32	100.25	100.05	100.04	100.23	100.84	100.24	100.44	99.98	100.44

3.3. Chemical composition

Al_2O_3 occurs in higher amounts in all samples except of F6 (wood chips) and in all samples from the second cyclone (Table 3). This is also in accordance with literature review [5]. The addition of coal in the fuel blend enriches the ash composition in Fe. Oxides containing this element (Fe_2O_3) were also identified by means of XRD. This oxide was present as hematite or maghemite in almost all samples examined, except of F6 (wood chips) and C6 (wood chips), which contained approximately only 2.5% of Fe_2O_3. On the other hand wood fly ash is rich in phosphorous, as seen in the aforementioned tables. This phosphorous originates from the wood chips used [5]. Consequently, P_2O_5 was significant enriched in the fly ash samples originating from wood chips. Moreover, calcium is also an important component in wood. As expected, the Ca levels of fly ash from co-combustion of wood with coal were lower than those of fly ash coming from the combustion of wood. It is generally noticed that when the proportion of wood in the coal/wood mixture is increased, the concentration of alkalis (Na and especially K) is also elevated.

In Table 4 the trace element concentrations determined by ICP-OES are presented.

Compared to coal ash from fluidised bed combustion described in the literature [5], the levels of As and Cr in many of the ashes included in this study seem rather low, especially the As content.

The concentration of As in lignite fly ash is attributed to the soil parent material, especially the earth crust which is enriched in this element. In these samples the elevated amounts of As could be due to the contact of fly ash with flue gases, the hydraulic conductivity (low in fly ash) and the surface area.

It appears that [6] most of the As in coal is associated with pyrite. In the majority of coals studied, As is primarily associated with massive or late-stage pyrite. In some cases As is associated with fine-grained pyrite and other sulphides.

Furthermore by increasing O_2 partial pressure, coal particles' combustion temperature can be raised. Thus Cr concentration is increasing by increasing O_2 concentration [7]. In addition boiler configuration and operation affect the behaviour of Cr. Lower emissions of Cr are usually detected at lower load [8]. The relatively high concentration of Cr could also be due to the attrition of the power plant equipment and the metallic nozzle of the probe used for sampling. Thus the presence of Cr could be indicative of contamination in association with stainless steel [9,10].

Table 4
Concentration (mg/kg) of trace elements in the fly ash

Sample	As	Cr	Co
F1	36 ± 3.5	190 ± 17	31 ± 3
F2	43 ± 4	182 ± 16	43.6 ± 4
F3	20 ± 2	180 ± 16	48 ± 4
F4	<5	100 ± 8	42.5 ± 4
F5	<5	115 ± 9	39.5 ± 4
F6	<5	29.5 ± 3	7.4 ± 0.7
AFBC ash residues form coal combustion [5]	10–300	1–500	–
C1	<5	160 ± 13	23 ± 2
C2	<5	145 ± 15	18 ± 2
C3	<5	180 ± 16	22 ± 2
C4	<5	32 ± 3	8.2 ± 0.8
C5	<5	72 ± 7	8.7 ± 0.8
C6	<5	20 ± 2	<5
AFBC ash residues form coal combustion [5]	10–300	1–500	–

A potential benefit [11] of fly ash as a soil amendment is as a source of nutrient elements for plant growth. Of the macronutrients (nitrogen, phosphorus, potassium, calcium, magnesium and sulphur), fly ash can supply sulphur, especially if it comes from the combustion of wood chips.

Besides the geochemical characteristics of trace element distribution in coals, emission of trace elements into the gaseous phase considerably depends on the condition of coal combustion (especially temperature) and effectiveness of the cleaning systems of the gases emitted [12,13]. It should be noted that most of the trace elements present in gaseous emissions are not in a true gas state, but are retained on the surface of solid particles (in ash). As a result, many elements end up in silicate melts or condense onto particle surface and unburned carbon (char); only a few trace elements are likely emitted into the atmosphere as gaseous compounds [12,14].

Furthermore, the distribution and concentration of trace elements in coal ashes as well as the mechanism of their enrichment are also closely related to furnace type and combustion process [12,15]. That is mainly because the combustion furnace type and combustion process are closely related to the degree of coal combustion, combustion temperature and combustion efficiency. If coal is efficiently combusted in the furnace, the carbon content in fly ash will be low; hence some of the elements associated with organic matter will be released into the flue gas [12].

3.4. Particle size distribution

Particle size distribution [2] of fly ashes is of particular interest when the material is used as a substitute for a portion of cement in concrete and in soil-stabilisation mixtures. Potential effects influenced by particle size distribution are related to: rheology, floe structure of the cement paste, permeability of the hard paste and rate of strength gain [16]. Particle characterization of fly ashes is also important for the recovery of metals. Leachability of the smallest particles is easier as they are more reactive, and in addition trace elements tend to concentrate in the finest particles [12]. Taking into account that the upgrading of fly ashes has economic limitations, a good characterization of the particle is essential to the process of metal recovery.

The particle size distribution of fly ashes is usually between 1 μm and 200 μm although coarser particles may exist especially in ashes from lignite [17]. Moreover, according to the literature [18], the fly ashes derived from CFB technologies contain smaller particles than the fly ashes coming from conventional combustion technologies.

Figs. 2 and 3 present the mean particle size distribution of the filter and second cyclone samples. The results show that the particle size distribution of the fly ashes ranges between 1 μm and 200 μm. Moreover a second peak occurs in the size distribution of all samples, in the range between 0.1 and 1 μm.

Fig. 2 depicts the particle size distribution of ashes from the fabric filter. As it can be seen samples F1 (lignite), F2 (wood chips/coal), F3 (wood chips) have a coarser particle size than F4 (coal), F5 (wood chips/coal) and F6 (wood chips). This result is due to the different combustion temperatures applied (Table 2) and the different fuel blends used.

The particle size in fly ash from a blend containing wood chips decreases with increasing percentage of wood chips. This is possibly due to the ease of combusting wood chips (Fig. 3).

Fly ash contains most elements found in the periodical table [19] and a number of researchers agree that many of the trace elements in fly ash show a definite concentration trend with decreasing particle size.

Concentrations of arsenic, cadmium, copper, gallium, molybdenum, lead, sulphur, antimony, selenium, thallium and zinc have been reported to increase with decreasing particle size [9]. This is evident by comparing the elemental content of fly ash collected from fabric filters to that of cyclones (Table 4).

3.5. X-ray diffraction (XRD)

Evaluating the XRD patterns, several distinct peaks were observed and the following minerals identified: quartz, hematite anhydrite and periclase. Some additional peaks on the diffractograms were attributed to maghemite, portlandite, gehlenite, magnesioferrite, lime, sylvite and rutile, which occurred in some samples.

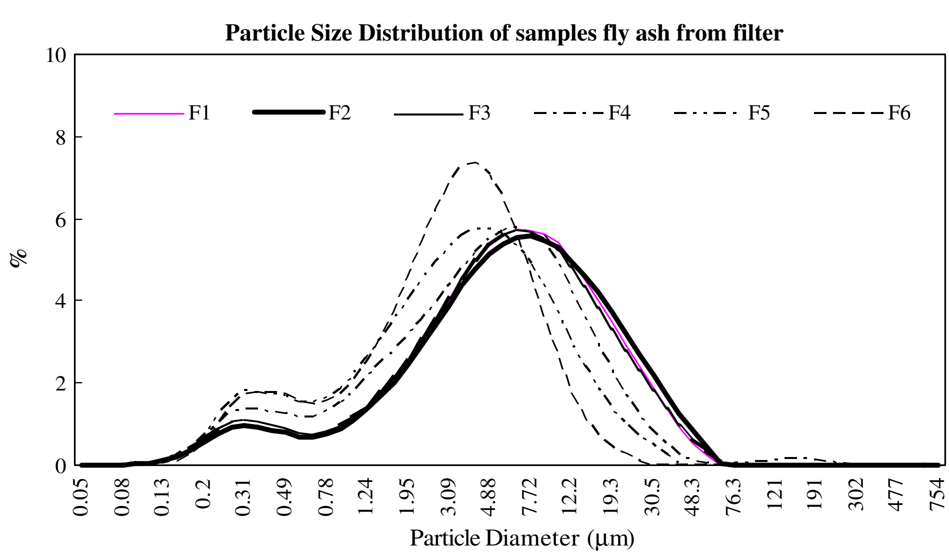

Fig. 2. Particle size distribution of samples from filter.

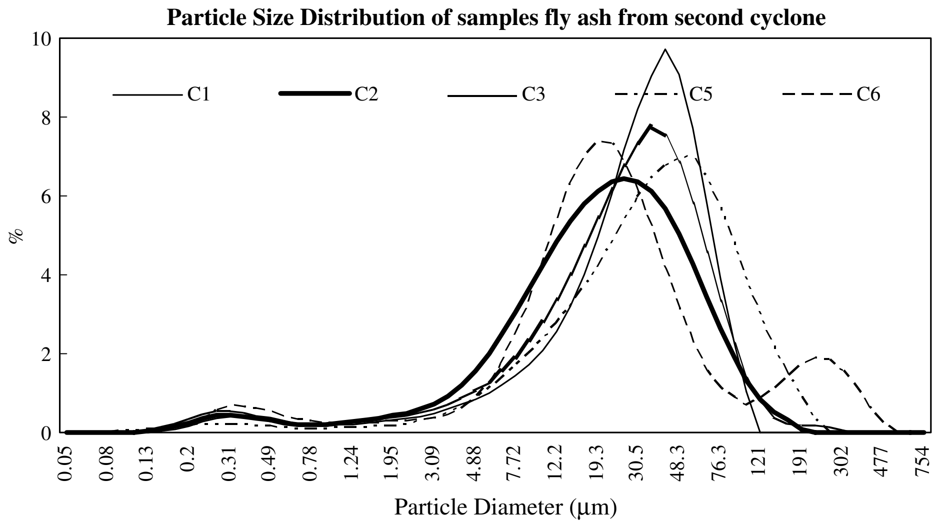

Fig. 3. Particle size distribution of samples from second cyclone.

The results of the mineralogical composition of these samples, obtained from the X-ray patterns, are presented in Table 5.

3.6. Phase-mineral composition and genetic features

Almost all ash samples contain quartz, which is a mineral that can be considered as primary, especially in cases where ash samples originate from fuels such as bituminous coal and lignite, which contain high amounts of silicon dioxide (Table 3).

The fly ashes from high sulphur coals have higher amounts of iron-bearing minerals compared to the fly ash from low sulphur coals. Formation of iron oxides is partially due to the decomposition of pyrite, which increases the SO_x content of the system and provides the basic cation for nucleation of hematite/magnetite, which then leads to the formation of magnetite microspheres [17].

Except of quartz, the majority of the samples contain hematite, especially in cases where bituminous coal is used. This type of coal has higher percentages of ferric oxide in comparison with lignite and the wood chips used (Table 3). Hematite was frequently accompanied by anhydrite, which probably forms from reactions between pyrite sulphur and calcite. According to literature review it was found that anhydrite and hematite began to form at 500 °C and were major phases through 800 °C [20].

Anhydrite is a mineral that is formed by dehydration of bassanite ($CaSO_4 \cdot 1/2H_2O$) or formed by calcite–pyrite reactions at temperatures higher than 500 °C [21]. It is stable at temperatures up to 990 °C.

Bassanite, which is formed by dehydration of original gypsum, is another important source of calcium in the system. Bassanite is stable at 400 °C, and in rare cases traces of it persist at 500 °C. During dehydration, which begins at 400 °C in some cases, anhydrite develops. This is a major phase through 1000 °C in some cases [20].

Maghemite is an oxidation product of magnetite (Fe_2O_3), an inverse spinel structure mineral and the thermal decomposition product of pyrite in coal [22]. It can be found almost in all samples blended with lignite.

Portlandite appears at low or medium combustion temperatures, by calcite heating [23]. Only the wood chips samples F6 (wood chips), C3 (wood chips) and C6 (wood

Table 5
Mineral phases in the fly ash samples

Mineral phases	F1	F2	F3	F4	F5	F6	C1	C2	C3	C4	C5	C6
Quartz – SiO_2	X	X	X	X			X	X	X	X	X	X
Maghemite – Fe_2O_3		X			X		X					
Hematite – Fe_2O_3	X	X	X				X	X	X	X	X	
Anhydrite – $CaSO_4$	X	X	X	X			X	X	X	X	X	
Periclase – MgO			X	X	X	X		X	X	X	X	X
Portlandite – $Ca(OH)_2$						X			X			X
Gehlenite – $Ca_2Al(Al,Si)_2O_7$					X							
Magnesioferrite – $MgFe_2O_4$				X								
Lime – CaO						X					X	
Sylvite – KCl						X						
Rutile – TiO_2	X											

"X" denotes that the corresponding crystalline phase was identified using XRD.

chips) include this mineral. In general, wood chips are abundant in calcium and probably containing calcite.

Periclase [21] is a secondary mineral and it is formed from the decomposition of dolomite ($CaMg(CO_3)_2$) through magnesite. It does not appear in the samples which originate from lignite (F1, C1).

The presence of hematite in combination with the presence of periclase leads to the formation of magnesioferrite ($MgFe_2O_4$). This mineral appears in sample F4 (coal). So, the absence of hematite may be due to its participation in the formation of magnesioferrite. On the other hand the presence of periclase in the same sample may be due to the high amount of this mineral which did not fully participate in the formation of magnesioferrite.

Lime forms due to the decomposition of calcite or from dolomite through calcite [21] and occurs only in sample C5 (coal/wood chips).

Gehlenite originates from carbonate–silicate-spinel reactions and appears only in sample F5 (coal/wood chips). This result leads to the assumption that calcium from wood chips and silicon dioxide from coal in this proportion is able to form this mineral.

Sylvite appears only in sample F6 (wood chips) and is possibly formed due to the high alkaline content of wood chips.

4. Conclusions

Compared to fly ash from combustion of wood chips, the co-combustion ashes studied had lower contents of calcium, potassium and magnesium. The blend of coal and wood chips adds aluminium and iron to the ash and the trace element levels were generally in the same ranges as for wood fly ash. The trace elements which may cause some concern are As and Cr in the case of sample F1 (lignite) and F2 (lignite/wood chips), which in addition are of small particle size. As aforementioned, many of the trace elements in fly ash show a definite concentration trend with decreasing particle size.

Fly ash deriving from the combustion of pure lignite and pure wood chips has a low LOI level. If the SO_3 content is taken additionally into account ($SO_3 < 5\%$), then the most attractive fly ash to the cement industry is C1 (lignite), collected at the second cyclone.

Moreover, the particle size distribution of the examined fly ashes is related to the fuels burned. The higher the percentage of biomass in the blends, the smaller the ash particle size.

The mineralogical composition of the fly ashes depends on the components of the blends and the combustion temperature. The presence of minerals such as anhydrite and portlandite is more possible when the used fuels contain calcium; therefore fly ash derived from wood chips contains the aforementioned minerals. Also, the occurrence of minerals such as sylvite is noticed when the fly ash derives from the combustion of biomass, due to the higher concentration of alkalis in this fuel.

The chemical characteristics of the blend of biomass-coals depend on the system of combustion and the fly ash capturing mechanism. That is mainly because the combustion furnace type and combustion process are closely related to the degree of coal combustion, combustion temperature and combustion efficiency.

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References

[1] European Coal Combustion Products Association e.V.

[2] Gutierrez B, Pazos C, Coca J. Characterization and leaching of coal fly ash. Waste Manage Res 1993;11:279–86.

[3] Standard specification for fly ash and raw calcined natural pozzolan for use as a mineral admixture in Portland cement concrete, ASTM C 618-94a. Philadelphia: American Society for Testing and Materials; 1995.

[4] Foner HA, Robl TL, Hower JC, Graham UM. Characterization of fly ash from Israel with reference to its possible utilization. Fuel 1999;78:215–23.

[5] Steenari BM, Lindqvist O. Fly ash characteristics in co-combustion of wood with coal, oil or peat. Fuel 1999;78:479–88.

[6] Finkelman RB. Modes of occurrence of potentially hazardous elements in coal: levels of confidence. Fuel Process Technol 1994;39:21–34.

[7] Helble JJ. Trace element behaviour during coal combustion: results of a laboratory study. Fuel Process Technol 1994;39(1–3):159–72.

[8] Danihelka P, Ochodek T, Noskievi P, Seidlerova J. Heavy metals and coal combustion. Proceedings of the 23rd international technical conference on coal utilisation and fuel systems, Clearwater, FL, USA, 9–13 March 1998. Washington, DC, USA: Coal and Slurry Technology Association; 1998. p. 977–86 [March 1999].

[9] Goodarzi F. Morphology and chemistry of fine particles emitted from a Canadian coal-fired power plant. Fuel 2006;85:273–80.

[10] Goodarzi F, Huggins FE. Speciation of chromium in feed coals and ash byproducts from canadian power plants burning subbituminous and bituminous coals. Energy Fuels 2005;19:2500–8.

[11] Strock NG, Stehouwer RC. Soil additives and soil amendments, Chapter 10. In: White WB, editor. Coal ash beneficial use in mine reclamation and mine drainage remediation in Pennsylvania. Harrisburg, PA; 2004. p. 302–15.

[12] Liu G, Zhang H, Gao L, Zheng L, Peng Z. Petrological and mineralogical characterizations and chemical composition of coal ashes from power plants in Yanzhou mining district, China. Fuel Process Technol 2004;85:1635–46.

[13] Meij R. The distribution of trace elements during the combustion of coal. In: Swaine DJ, Goodarzi F, editors. Environmental aspects of trace elements in coal. Dordrecht, The Netherlands: Kluwer Academic Publishers; 1995. p. 111–27.

[14] Maroto-Valer MM, Taulbee DN, Hower JC. Characterization of differing forms of unburned carbon present in fly ash separated by density gradient centrifugation. Fuel 2001;80:795–800.

[15] Clarke LB. The fate of trace elements in emissions control systems. In: Swaine DJ, Goodarzi F, editors. Environmental aspects of trace elements in coal. Dordrecht, The Netherlands: Kluwer Academic Publishers; 1995. p. 128–45.
[16] Diamond S. Rapid particle size analysis of fly ash with a commercial laser diffraction instrument. Mater Res Soc Symp Proc 1988;113:119–27.
[17] Goodarzi F. Characteristics and composition of fly ash from Canadian coal-fired power plants. Fuel 2006;85:1418–27.
[18] Jankowski J, Ward CR, French D, Groves S. Mobility of trace elements from selected Australian fly ashes and its potential impact on aquatic ecosystems. Fuel 2006;85:243–56.
[19] Swaine DJ. Trace elements in coal. London: Butterworth; 1990. 292 p.
[20] Mitchell RS, Gluskoter HJ. Mineralogy of ash of some American coals: variations with temperature and source. Fuel 1976;55:90.
[21] Filippidis A, Georgakopoulos A, Kassoli-Fournaraki A. Mineralogical components from ashing at 600 °C to 1000 °C of the Ptolemais lignite, Greece. Trends Miner 1992;1:295–300.
[22] Galbreath KC, Zygarlicke CJ, Tibbetts JE, Schulz RL, Dunham GE. Effects of NO_x, α-Fe_2O_3, γ-Fe_2O_3, and HCl on mercury transformations in a 7-kW coal combustion system. Fuel Process Technol 2005;86(4):429–48.
[23] Filippidis A, Georgakopoulos A. Mineralogical and chemical investigation of fly ash from the main and northern lignite fields in Ptolemais, Greece. Fuel 1992;71:373–6.